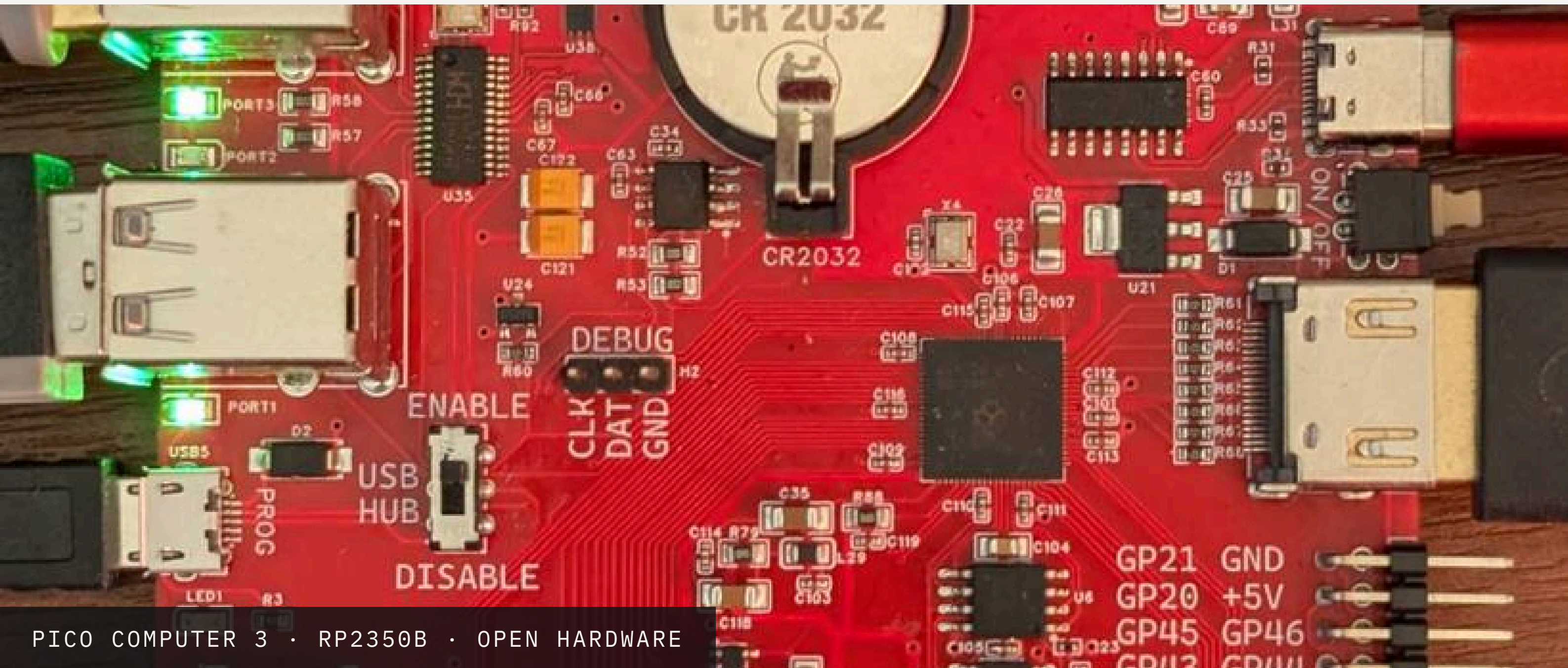
Unix, self-hosted.

A genuine multi-tasking Unix on a single RP2350 microcontroller — booting from SD card, driving an HDMI screen and a USB keyboard, and carrying a complete C compiler that runs **on the machine itself**. No cross-development. No PC in the loop.



PICO COMPUTER 3 · RP2350B · OPEN HARDWARE

375 MHz	**8** MB	**64**	**80×40**
Dual Cortex-M33	PSRAM for heap & swap	Concurrent processes	Colour console on HDMI

A real Unix

Pre-emptive multitasking, pipes, a Bourne shell and the classic V7 userland — plus the one true `awk`, `vi` and man pages.

It compiles itself

A complete C89 toolchain on the board, generating native ARM code. Write it, build it and run it without leaving the machine.

Two BASICs as well

R. T. Russell's BBC BASIC with graphics and sound, and an MMBasic translator that compiles to native code.

The only Pico that builds its own programs.

Everything below happens at the # prompt on the board. Nothing is cross-compiled, downloaded or flashed — the editor, the compiler, the runtime and the program all live on the SD card.

```
login: root
# cc bench.bas
wrote bench.mb.c
.....................................
# ./bench.bc
MMBASIC benchmark (C) KnivD 2016
Performance:    55353 grains
```

A published MMBasic benchmark, unmodified. The same program under the MMBasic interpreter, on the same board at the same clock, scores about 12,000.

A C compiler, not a cross-compiler

`cc` is a complete C89 toolchain running on the hardware. It translates each function into native Thumb-2 for the Cortex-M33, so compiled programs run as machine code — not as an interpreted intermediate.

167,000 Dhry/s

Dhrystone 2.1, compiled on the machine — 44% of the same benchmark cross-compiled by `gcc -O2` for the same chip.

Its bytecode is language-neutral by design, so other front ends can share the back end — `mmbc`, the MMBasic translator, is the first.

A 3,200-line astronomical calculation

SOLAR_ECLIPSE.BAS

MMBasic interpreter	12.5 s
MicroPython	8.8 s
Fuzix — mmbc + cc	**2.17 s**

Same source, same board, same clock. Every digit of the printed answer is identical in all three — the speed is the only thing that changes.

THREE LANGUAGES, ONE MACHINE

C

Full C89 bar bitfields, doubles on the RP2350 co-processor, and **31 board headers** — graphics, pins, I²C, SPI, sound.

`cc prog.c`

MMBasic

One command: `mmbc` translates it to C and `cc` compiles it native. 220 of MMBasic's 353 keywords.

`cc prog.bas`

BBC BASIC

R. T. Russell's own. MODE 0–5, four-channel sound with ENVELOPE, ADVAL joysticks, inline assembler.

`bbcbasic`

And a complete ANS Forth, `fforth`, because why not

Specifications, and what's on the card.

HARDWARE & KERNEL

Processor	RP2350B, dual Cortex-M33 at 375 MHz
Memory	340 KB of program RAM in 4 KB pages — 64 processes, or ~292 KB for one
PSRAM	8 MB, holding program heaps and swapped-out processes
Scheduling	Pre-emptive — a runaway program can always be stopped
Display	HDMI at 640×480 — an 80×40 colour ANSI console, mirrored to USB-C serial
Keyboard	USB HID, six layouts: US, UK, DE, FR, ES, BE
Storage	SD card root on FS32 — 32-bit block numbers, so up to 2 TB. 15 MB of flash holds a recovery system
Audio	PCM5102 I²S DAC to a 3.5 mm jack
Clock	DS3231, battery-backed, read and set from the shell
Other I/O	Second serial port, QWIIC I²C, four joystick inputs, four ADC channels, PWM and servo

ON THE SD CARD

The V7 userland
Bourne `sh` and the classic set — ls, cp, grep, sed, find, sort, tar, dd, ps, cron and the rest.

The one true awk
Lucent's, by one of the original authors. Associative arrays, functions, getline — not a subset.

Three editors
`mmedit`, MMBasic's own full-screen editor with keyword colouring; `vi`; and V7's `ed`.

Headers for the board
One per primitive — graphics, pins, I²C, SPI, one-wire, sprites, sound — so a program carries only what it names, plus `pc3io.h` for the pin and ADC registers.

Media
MP3, WAV, FLAC and MOD playback; JPEG, PNG and BMP loading; sprites and an off-screen framebuffer.

Manual pages, and a cave
Plus a FAT partition a PC can drop files onto, and `advent` — the original Colossal Cave.

WHAT IT IS NOT

There is no networking, so nothing to browse and nowhere to `ssh` to. `cc` has no linker or assembler, so a program is one source file, and bitfields are the one piece of C89 missing. The V7 tools are period-correct where it shows: `grep` has no `-E`, `diff` has no `-u`. All of it is in the manual rather than discovered the hard way.

GETTING IT RUNNING

1. Hold BOOT, click RESET, and drag `fuzix.uf2` onto the drive that appears.
2. Write the ready-made card image to a 1 GB or larger SD card with any raw-image tool.
3. Plug in a monitor and a USB keyboard and switch on. At the `bootdev:` prompt, type `hdb2` — the root partition on the card.
4. Log in as `root`. There is no password.

All of it is open source.

The kernel, the compiler, the userland, both manuals and the board's own design files. Build it from a clean clone, or write the image and go.

Kernel and the PC3 port — branch pc3
`github.com/UKTailwind/FUZIX`

PCB design files for the board
`github.com/UKTailwind/Pico-Computer-3`

MicroPython for the same board
`github.com/UKTailwind/micropython`

MMBasic firmware
`github.com/UKTailwind/PicoMite`

Fuzix by Alan Cox · BBC BASIC by R. T. Russell · MMBasic by Geoff Graham and Peter Mather · Pico Computer 3 hardware and this port by Peter Mather